

SymPareto: A Dynamic Gain–Degradation Co-Evolution Framework for Pareto-Optimal Task Scheduling in Autonomous Intelligent Systems

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Abstract—Artificial intelligence (AI), particularly advanced machine learning algorithms and autonomous systems, is rapidly transforming task structures across industries, blurring the lines between human and machine roles. As AI expands its scope, concerns arise about the potential erosion of human skills, especially as task niches narrow and are reallocated to increasingly capable machines. To address this, we propose SymPareto, a feedback-driven framework that models and optimizes the co-evolutionary dynamics between humans and AI agents. SymPareto introduces a novel approach by treating task scheduling as an ecological intervention, simultaneously guiding human skill retention (modeled via a “use-it-or-lose-it” dynamic) and AI capability growth (modeled through a “reproduction–diffusion” process in task space). This leads to an asymmetric yet synergistic evolution of task niches, similar to species proliferation and boundary negotiation in ecological systems. The framework integrates this dynamic task ecology with a bi-objective Pareto optimization process, searching for optimal trade-offs between human skill utility and AI task utility. Compared to non-Pareto optimization baselines, SymPareto achieves a 77.18% improvement in performance. This approach guides the system towards a stable Pareto frontier of co-adaptive task allocations. We implement SymPareto on a simulation platform built from the real-world smart home dataset InteractMine, evaluating its performance under various

scheduling strategies. Results show that SymPareto significantly improves human skill retention, promotes broader AI capability diffusion, and dynamically stabilizes the boundaries between human and AI tasks. Beyond technical efficacy, SymPareto offers a fresh perspective on sustainable symbiotic intelligence. It moves beyond the dichotomy of technological substitution versus human marginalization, providing a principled, interpretable framework for designing stable, fair, and mutually beneficial human-AI ecosystems.

Index Terms—Human–AI co-evolution, skill boundary reconstruction, symbiotic intelligence, task allocation, Pareto optimization.

I. INTRODUCTION

THE rapid advancement of artificial intelligence (AI) is fundamentally reshaping task ecosystems across diverse sectors, from smart homes and healthcare to industrial manufacturing [1]. As AI expands its capabilities, it continuously redefines human–machine roles, frequently encroaching on—and in some cases replacing—human-dominated task niches [2], [3]. This transformation demands an urgent rethinking of how humans and machines coexist and collaborate [4], [5]. Drawing an analogy to ecological systems, where species competition and succession govern survival, AI’s encroachment raises critical questions: will humans face a “use-it-or-lose-it” crisis, as their task niches contract and skills atrophy? Can we design a system that fosters long-term, sustainable human–AI symbiosis, ensuring mutual benefits rather than one-sided displacement?

Existing research on human–machine collaboration has predominantly focused on static task allocation and short-term efficiency gains [6], [7], overlooking the long-term co-evolutionary dynamics between human and AI skills under changing task ecosystems. Traditional views tend to treat AI’s expansion as a simple species-replacement phenomenon, a “substitution effect” [8], rather than acknowledging the potential for co-evolutionary synergy [9]. In reality, as task boundaries shift in response to evolving AI capabilities, these boundaries should dynamically adapt in a manner that fosters mutual growth, creating co-evolutionary niches governed by Pareto-optimal principles.

In response to these challenges, this paper introduces SymPareto, a novel feedback-driven framework that models the intertwined evolution of human and AI capabilities.

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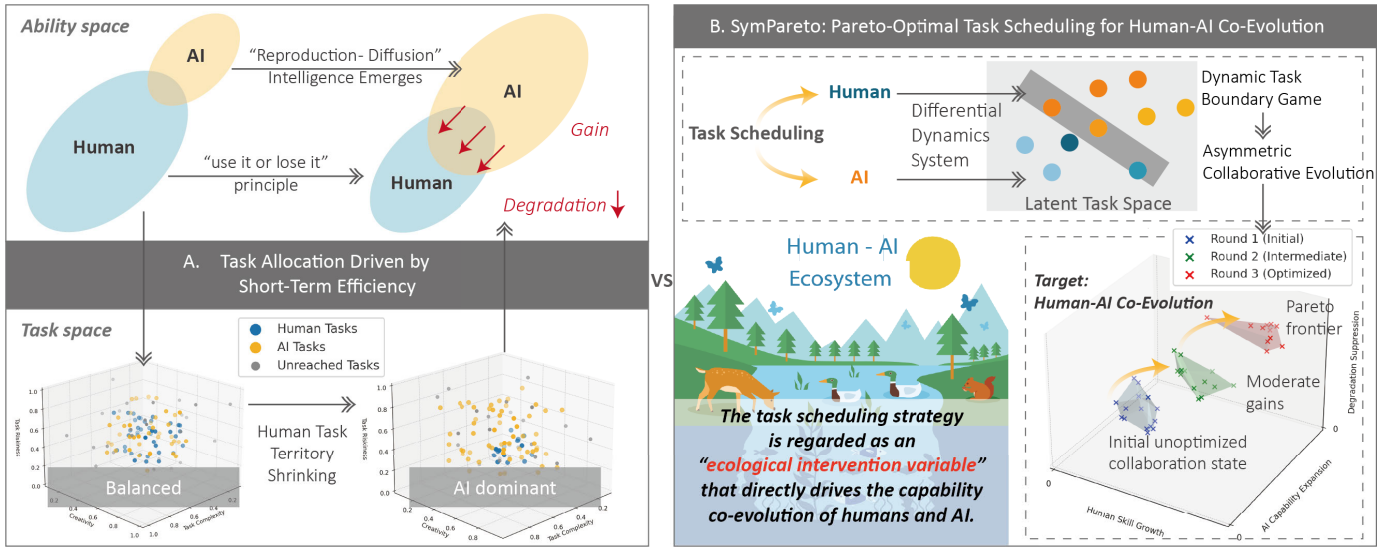


Fig. 1. **Task allocation paradigms and their effects on human–AI skill co-evolution.** (A) Conventional allocation strategies prioritize short-term efficiency, often inducing a competitive dynamic between human and AI agents. As AI capabilities expand via mechanisms of reproduction and diffusion, human skills may atrophy under a “use-it-or-lose-it” dynamic. This asymmetry progressively compresses the human task niche, resulting in an AI-dominated task ecology. (B) The proposed SymPareto framework introduces a Pareto-optimal task scheduling approach grounded in differential gain–degradation dynamics. By treating task allocation as an ecological intervention variable, the system enables asymmetric but synergistic evolution, sustaining mutual growth. Iterative optimization steers the task distribution toward the Pareto frontier, preventing unilateral dominance and fostering a balanced, co-adaptive human–AI ecosystem.

Unlike prior approaches, SymPareto treats task scheduling as an ecological intervention variable—a dynamic force that drives both human skill retention, embodied through a “use-it-or-lose-it” principle, and AI capability growth, modeled as a “reproduction–diffusion” process across task spaces. This model enables task boundaries to evolve continuously through an iterative negotiation between humans and AI, mirroring the dynamics of species expansion and ecological competition. Figure 1 provides a visual overview contrasting conventional task allocation approaches with the proposed SymPareto framework. As shown in Panel A, short-term efficiency-driven strategies often lead to skill degradation and AI dominance, shrinking the human task territory. In contrast, Panel B illustrates how SymPareto dynamically balances gain and degradation through Pareto-optimal scheduling embedded within a differential dynamics system, enabling long-term co-evolution and mutual benefit across a latent task space.

A key innovation of our framework is its tight integration with Pareto optimization, which distinguishes it from previous approaches that typically aim for a single, static trade-off point or rely on fixed reward structures. SymPareto formulates task scheduling as a bi-objective optimization problem that simultaneously maximizes both human skill utility and AI capability utility. This dual-objective approach enables the system to dynamically explore and converge towards a set of non-dominated solutions along the Pareto frontier, ensuring a long-term balance between human and AI growth without compromising either party’s potential. Through this framework, co-evolution becomes not merely a byproduct of task allocation, but an explicit optimization target, driving the system toward sustained and synergistic human–AI collaboration.

To realize this vision, we embed SymPareto within a simulation platform that leverages the real-world InteractMine dataset, capturing diverse human–AI interactions in smart home environments. This allows us to evaluate the effects of

various ecological scheduling strategies on key system properties, such as human skill retention, AI diffusion trajectories, and ecological niche stability over time. The experiments aim to determine whether the system can prevent unilateral dominance by either human or AI agents, foster sustained collaboration, and adaptively adjust task boundaries to maintain a balanced human–AI ecosystem.

The framework incorporates the following core mechanisms:

- **Dynamic ecological modeling of human skills:** Human skill evolution is modeled as a niche-occupancy process, where capabilities follow nonlinear differential dynamics based on the “active growth–passive decay” principle, embodying the ecological “use-it-or-lose-it” concept.
- **AI capability proliferation and diffusion mechanism:** The AI task-expansion process is likened to species proliferation in an ecosystem, where the diffusion of AI capabilities across tasks is driven by both task participation shares and cross-task transmission effects.
- **Multi-objective dynamic regulation of ecological niches:** The framework employs a human–machine task allocation matrix and collaboration coupling intensity to dynamically regulate the distribution of task niches and matching capabilities, optimizing the system’s collective progression along the Pareto frontier.
- **Validation of ecological co-evolutionary strategies:** The NSGA-II multi-objective evolutionary algorithm is utilized to continually search for and compare the effects of different scheduling strategies, assessing the system’s long-term stability, skill-level retention, and AI expansion potential.

Compared with traditional static-partitioning or reinforcement-learning-based approaches, the SymPareto framework offers three core advantages: (i) it interprets task-space and capability evolution through explicit ecological-dynamics mechanisms, providing greater theoretical

interpretability and academic rigor; (ii) its model parameters are well-defined and control variables clear, facilitating systematic scheduling-strategy design; and (iii) it supports real-data integration, ensuring strong simulation reproducibility and ecologically grounded optimization interpretability.

At the experimental level, we build a simulation platform based on the real-world smart home dataset **InteractMine** to systematically analyze human skill retention, AI capability diffusion pathways, and long-term system stability under different ecological strategies. The results demonstrate that the **SymPareto** framework effectively enables the co-evolution of human–AI niches and dynamic optimization of task boundaries, contributing to sustained win–win balance within the task ecosystem.

Our main contributions are summarized as follows:

- We introduce **SymPareto**, a novel feedback-driven differential dynamics framework that explicitly models the co-evolutionary process between human skill retention, guided by the “use-it-or-lose-it” principle, and AI capability growth through a structured “reproduction–diffusion” mechanism in task spaces.
- We formulate task scheduling as dynamic ecological interventions, embedding scheduling decisions within a unified differential system. This integration allows systematic exploration and optimization of human–AI interaction boundaries through a Pareto-optimization framework powered by the NSGA-II algorithm.
- We comprehensively evaluate SymPareto on simulations built from the real-world *InteractMine* smart-home dataset, systematically demonstrating its efficacy in preserving human skills, expanding AI capability coverage, and dynamically stabilizing task boundaries. Compared to static and myopic baselines, our results provide clear evidence that SymPareto significantly enhances sustainable human–AI collaboration.

The remainder of the paper is organized as follows. Section II reviews related work on human–AI collaboration and multi-objective optimization frameworks. Section III presents the details of the **SymPareto** framework, including human skill dynamics modeling, AI capability growth, and task-scheduling optimization via NSGA-II. Section IV describes the experimental design and provides a comprehensive analysis of results on the *InteractMine* dataset. Section V discusses the scalability, ethical considerations, and broader applicability of the proposed framework. Finally, Section VI summarizes the key findings and outlines future directions for advancing sustainable human–AI co-evolution.

II. RELATED WORK

Research on human–AI teaming in human–computer interaction typically focuses on short-term optimization of performance-related metrics—such as interface adaptability, task accuracy, and cognitive load—through mechanisms like adaptive automation and mixed-initiative control [10], [11], [12]. While these approaches improve task efficiency in the immediate term, they often treat humans as static components within the control loop, overlooking the long-term implications of automation on human skill development.

Recent studies have begun to question whether persistent

AI support leads to human deskilling, but these are typically empirical or observational in nature and lack systematic modeling of the underlying dynamics [13], [14], [15].

To balance competing objectives in joint human–AI systems, multi-objective optimization has emerged as a powerful paradigm. Evolutionary algorithms, particularly NSGA-II and its derivatives, are frequently employed to explore Pareto-efficient trade-offs between factors such as performance, energy consumption, and workload [16], [17], [18]. However, these applications often assume static utility functions or fixed task capabilities, failing to account for how human and AI capabilities co-evolve over time under different allocation strategies. Purely game-theoretic approaches [19], [20] go further in simplifying the interaction, casting human–AI collaboration as a zero-sum game with fixed pay-off structures. In such models, gains by AI are implicitly interpreted as losses for humans, reinforcing the notion of competitive displacement rather than cooperative adaptation.

More recent work has started to reframe the problem as dynamic frontier exploration in multi-agent systems, showing that Pareto-optimal trajectories can enable joint expansion of agent capabilities through adaptive reallocation [21], [22]. However, these frameworks often remain abstract, lacking concrete mechanisms to model the skill dynamics of individual human agents or the structural propagation of AI capabilities across heterogeneous task spaces. They also rarely incorporate real-world constraints such as skill degradation, substitution pressure, or task-specific learning plasticity.

In contrast to prior efforts, we propose **SymPareto**, a dynamic framework that couples individual-level human skill evolution with AI diffusion and feedback-driven task scheduling. By embedding utility trade-offs into an evolving differential system, SymPareto enables intervention in the co-evolution process to discover sustainable, Pareto-efficient collaboration trajectories.

III. METHODOLOGY

A. Problem Formulation and Notation

With the rapid development of artificial intelligence (AI), the boundaries between humans and AI in the task space are no longer static but are dynamically adjusted and continuously reconstructed. This dynamic boundary presents both challenges and opportunities for collaborative task allocation.

To address this, we propose a modeling framework named **SymPareto**, which formally describes the co-evolutionary process of humans and AI within a multi-task, multi-objective system. The goal is to drive the overall system along a *Pareto-optimal trajectory* without compromising the interests of either party.

Pareto Optimality is a fundamental concept in multi-objective optimization. It refers to a state where no objective can be improved without deteriorating at least one other. In this work, we consider human skill development and AI task capabilities as two primary optimization dimensions, aiming to maximize the system’s collaborative performance without diminishing the utility of either agent.

Figure 2 presents an overview of the proposed **SymPareto** framework, which integrates human skill dynamics, AI capability diffusion, and Pareto-optimal task scheduling within a

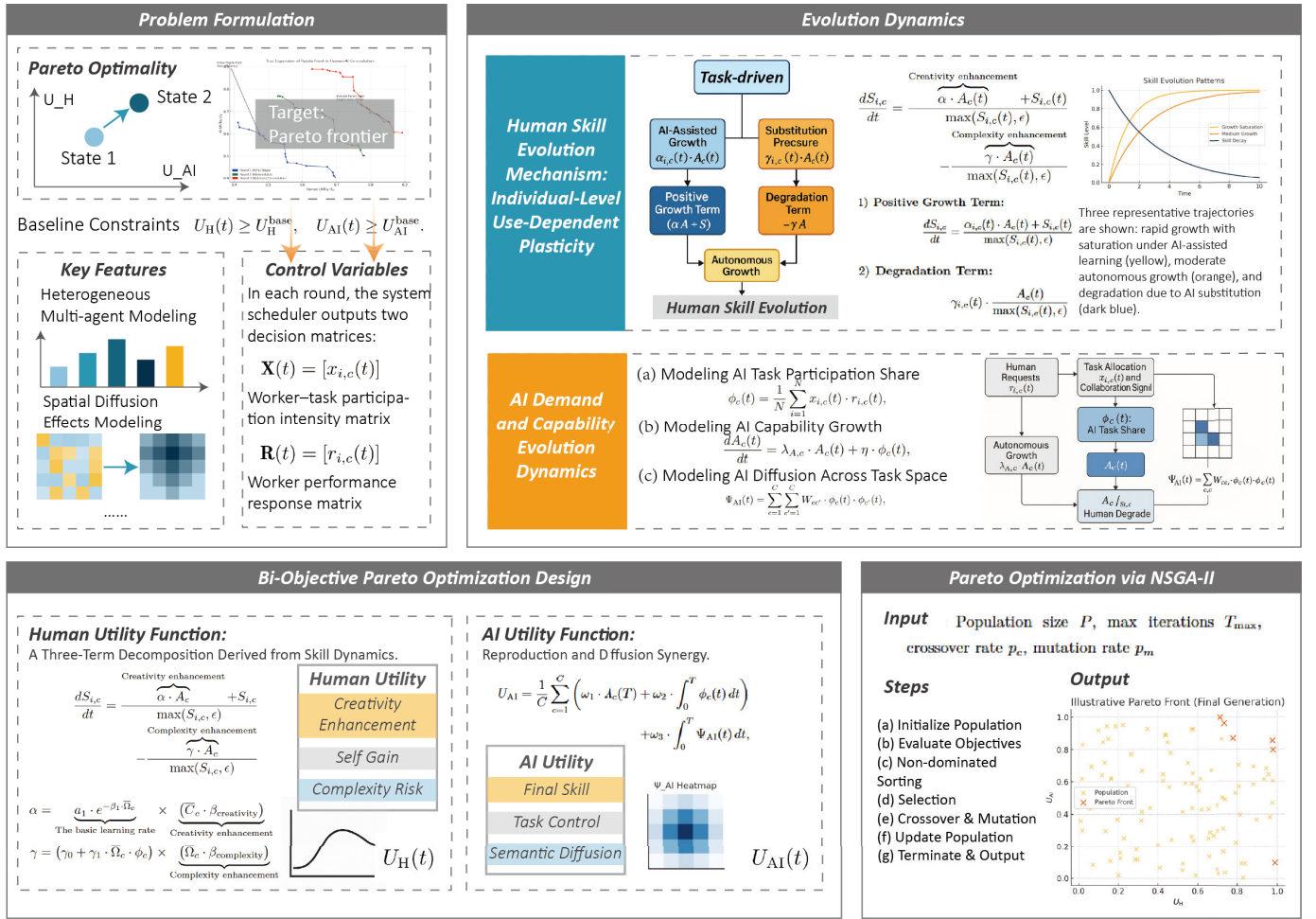


Fig. 2. **SymPareto framework for dynamic human-AI task co-evolution and Pareto-optimal scheduling.** The system models task allocation as an evolving interaction between humans and AI within a constrained ecology. Human skills adapt through use-dependent plasticity, while task boundaries shift in response to agent dynamics and performance feedback. Utility functions reflect distinct human and AI objectives, and multi-objective optimization is used to approach the Pareto frontier. The resulting scheduling strategy supports asymmetric collaboration, sustained skill growth, and avoidance of unilateral dominance.

unified differential system. The model formulates human-AI interaction as a co-evolutionary process: human skills evolve under use-dependent plasticity driven by task participation, while AI capabilities expand across the task space via a structured reproduction-diffusion mechanism. Utility functions are constructed to reflect both human-centered and AI-centered performance objectives, and a bi-objective optimization procedure based on NSGA-II is employed to guide the system toward Pareto-optimal configurations. This design enables dynamic niche boundary adjustment and long-term asymmetric collaboration under constrained ecological conditions.

This human-AI co-evolutionary modeling problem exhibits the following key features:

- **Multi-objective Collaborative Optimization:** The system jointly optimizes human utility U_H and AI utility U_{AI} by coordinating their task scheduling and interaction intensities through a Pareto optimization paradigm.
- **Heterogeneous Multi-agent Modeling:** The system includes N heterogeneous human agents, each differing in parameters such as learning rate and degradation sensitivity, interacting with a single AI system equipped

with evolutionary and diffusion capabilities, forming an asymmetric game structure.

- **Task Category Space and Dynamic Allocation:** Each task category $c \in \{1, \dots, C\}$ is embedded into a shared latent space. The individual task participation $x_{i,c}(t)$ and AI substitution ratio $r_{i,c}(t)$ jointly determine the actual contribution within each category, leading to a *dynamically adjustable task boundary*.
- **Strong Coupling Between Control and Feedback:** The scheduling strategies $\mathbf{X}(t)$ and $\mathbf{R}(t)$ not only influence human skill evolution (see Eq. (15)) but also affect future task participation and capability boundaries through AI substitution and diffusion, thereby forming a *high-dimensional nonlinear feedback system*.
- **Modeling of Spatial Diffusion Effects:** AI utility depends not only on terminal capabilities $A_c(T)$ and participation share $\phi_c(t)$ in each task category, but also on cross-category generalization ability $\Psi_{AI}(t)$, which reflects AI's control range and generalization potential in the latent space (see Eq. (17)).

The mathematical formulation of this problem is a dynamic system optimization with control variables $\mathbf{X}(t)$ and $\mathbf{R}(t)$,

TABLE I
SUMMARY OF KEY NOTATIONS AND DIMENSIONS

Symbol	Dimension / Range	Description
T_c	$\mathbb{R}^m$	Attribute vector of category c
$W_{cc'}$	$[0, 1]$	Inter-category similarity (transferability)
$S_{i,c}(t)$	$[0, 1]$	Skill level of human i in category c
$A_c(t)$	$\mathbb{R}_+$	AI capability in category c
$x_{i,c}(t)$	Decision variable	Human task allocation ratio
$r_{i,c}(t)$	Decision variable	AI coupling intensity
$\phi_c(t)$	Derived variable	AI participation share (replacement intensity)
U_H, U_{AI}	Objective functions	Human and AI system utility
$\bar{C}_c$	$\mathbb{R}_+$	Average creativity of category c
$\bar{\Omega}_c$	$\mathbb{R}_+$	Average complexity of category c
C, m, N, T	Scalars	#categories, attribute dimension, #humans, final time
α_1	Scalar	The efficiency coefficient of human learning assisted by AI
β_1	Scalar	The efficiency coefficient of human self-learning
γ_0	Scalar	The attenuation coefficient of basic skills
γ_1	Scalar	The attenuation coefficient induced by AI substitution
$\beta_{creativity}$	Scalar	Creativity enhancement coefficient
$\beta_{complexity}$	Scalar	Complexity enhancement coefficient
ε	Scalar	Stability constant, lower bound for S ($\approx 10^{-3}$)
θ	Vector	Strategy vector encoding $X(t)$ and $R(t)$

aiming to identify an evolutionary trajectory that achieves Pareto-optimality between human development and AI evolution (see Eq. (18)).

We will model the following components in subsequent sections:

- The human skill evolution mechanism in tasks, incorporating both positive growth and AI-induced degradation;
- The AI capability growth and its reproduction–diffusion dynamics in the task space;
- The control mechanism that dynamically schedules tasks and adjusts interaction strength based on system states;
- The definition of bi-agent Pareto objectives and the design of a feasible solving algorithm.

1) *Task Categories and Attributes*: Let the task space consist of C categories, denoted by $\mathcal{C} = \{1, \dots, C\}$. Each category c is associated with a static m -dimensional attribute vector:

$$T_c = [t_c^{(1)} \ t_c^{(2)} \ \dots \ t_c^{(m)}]^\top \in \mathbb{R}^m, \quad (1)$$

representing task requirements in terms of complexity, creativity, risk, etc. The inter-category transferability is captured by a propagation matrix $W \in [0, 1]^{C \times C}$. The W matrix is used to quantify the similarity between different tasks.

2) *Task Allocation Matrix*: Let $\mathbf{X}(t) = [x_{i,c}(t)] \in [0, 1]^{N \times C}$ denote the task allocation matrix, where $x_{i,c}(t)$ indicates the proportion of time that human i is allocated to task category c at time t . To ensure conservation of task assignments, a normalization constraint must be satisfied:

$$\sum_{c=1}^C x_{i,c}(t) = 1, \quad \forall i = 1, \dots, N, \quad \forall t \in [0, T]. \quad (2)$$

3) *AI Coupling Matrix*: Let $\mathbf{R}(t) = [r_{i,c}(t)] \in [r_{\min}, r_{\max}]^{N \times C}$ represent the coupling control matrix, where $r_{i,c}(t)$ denotes the intensity with which human i requests AI assistance in category c . This variable modulates the extent of AI involvement in task execution: higher values indicate greater reliance on automation. To avoid extreme dependency or complete disengagement, the coupling intensity is constrained

within a predefined range:

$$r_{i,c}(t) \in [r_{\min}, r_{\max}], \quad \forall i, c, t. \quad (3)$$

See Table I for a list of key symbols and their dimensions.

In the subsequent modeling, the AI replacement share $\phi_c(t)$ drives the dynamic degradation term $A_c(t)/S_{i,c}(t)$ in human skill evolution, thereby reflecting the asymmetric co-evolution mechanism in human–AI collaboration.

B. Human Skill Evolution Mechanism: Individual-Level Use-Dependent Plasticity

In human–AI collaborative task systems, the evolution of human skills is influenced not only by task engagement but also by the degree of AI substitution and the individual's learning state. To model this dynamic process more realistically, we introduce a differential equation-based *use-dependent plasticity* model, integrating three mechanisms—AI-assisted learning, autonomous growth, and substitution-induced degradation—to capture the asymmetric cognitive evolution of individuals under multi-task collaboration.

1) *Skill Evolution Equation*: Let $S_{i,c}(t) \in (0, 1]$ denote the skill level of human i in task category c at time t . Its evolution follows the differential equation:

$$\frac{dS_{i,c}}{dt} = \frac{\alpha \cdot A_c(t) + S_{i,c}(t)}{\max(S_{i,c}(t), \varepsilon)} - \frac{\gamma \cdot A_c(t)}{\max(S_{i,c}(t), \varepsilon)} \quad (4)$$

$$\alpha = a_1 \cdot e^{-\beta_1 \cdot \bar{\Omega}_c} \times (\bar{C}_c \cdot \beta_{creativity})$$

$$\gamma = (\gamma_0 + \gamma_1 \cdot \bar{\Omega}_c \cdot \phi_c) \times (\bar{\Omega}_c \cdot \beta_{complexity}) \quad (5)$$

Here, $\varepsilon \ll 1$ is a small constant used to prevent numerical instability as $S_{i,c}(t) \rightarrow 0$, ensuring simulation and integration stability. Where α is the learning efficiency coefficient modulating skill improvement rates, and γ is the skill degradation coefficient controlling skill diminishment speeds.

2) Structural Mechanism Analysis:

1) Positive Growth Term:

$$\frac{dS_{i,c}}{dt} = \frac{\alpha_{i,c}(t) \cdot A_c(t) + S_{i,c}(t)}{\max(S_{i,c}(t), \varepsilon)} \quad (6)$$

This term adopts a $1/S$ -type growth structure to reflect diminishing marginal returns: skill acquisition accelerates rapidly when the human skill level is low, then gradually saturates as proficiency increases. This is consistent with the fact that the integral of $1/S$ yields $\ln(x)$ —a function that grows quickly at first and then flattens out—capturing the intuition that humans learn rapidly at early stages but experience slower progress later. The growth is jointly driven by AI-provided coaching signals (via $\alpha_{i,c}(t)A_c(t)$) and intrinsic human motivation (via $\delta_{i,c}(t)$).

2) Degradation Term:

$$\gamma_{i,c}(t) \cdot \frac{A_c(t)}{\max(S_{i,c}(t), \epsilon)} \quad (7)$$

This term models cognitive degradation induced by AI substitution. It intensifies when the individual skill level is low ($S_{i,c}(t) \rightarrow 0$) and AI capability is high. The denominator includes $\epsilon \ll 1$ to prevent numerical instability as $S_{i,c}(t) \rightarrow 0$.

3) Asymmetric Feedback Dynamics:

The model inherently exhibits asymmetric feedback: growth decelerates as skills saturate ($S \rightarrow 1$), while degradation accelerates under low-skill conditions ($S \rightarrow 0$), highlighting the cognitive vulnerability to replacement.

This model offers several advantages:

- Compact structure with interpretable parameters, suitable for continuous-time control and optimization frameworks;
- Accurately reflects human cognitive changes under varying task engagement and AI substitution scenarios;
- Can be directly embedded into individual utility functions, system objectives, and optimal scheduling mechanisms;
- Supports multi-task generalization and heterogeneous population modeling, enabling system-level analysis of Pareto co-evolution.

In subsequent sections, we define both individual and population-level human utility functions based on this model (see Eq. (15)), which, together with AI objectives, form the bi-objective structure of the Pareto optimization framework.

C. AI Demand and Capability Evolution Dynamics

To mirror the modeling of human skill evolution, we formulate the behavior dynamics of AI in the system as a set of continuous-time differential equations, encompassing task participation, capability growth, and diffusion across the task space. Unlike human skill $S_{i,c}(t)$, which evolves through both growth and degradation, AI capability $A_c(t)$ generally follows a monotonic increase, with its task distribution driven by human demand intensity and task similarity.

1) Modeling AI Task Participation Share (Substitution Intensity): The task participation intensity of AI in category c —also interpretable as the degree of substitution—is determined by the aggregated task allocation and collaboration request from all human agents:

$$\phi_c(t) = \frac{1}{N} \sum_{i=1}^N x_{i,c}(t) \cdot r_{i,c}(t), \quad (8)$$

where $x_{i,c}(t)$ denotes the task allocation ratio of human i in category c , and $r_{i,c}(t)$ represents the intensity of collaboration request to AI. This variable reflects not only the extent of AI control over tasks but also serves as the core driver for human skill degradation via the term $\frac{A_c(t)}{S_{i,c}(t)}$.

2) Modeling AI Capability Growth: Reproduction-Driven Dynamics: The AI capability $A_c(t)$ in each task category increases over time, driven both by autonomous development and by a *reproductive effect* proportional to its task share $\phi_c(t)$:

$$\frac{dA_c(t)}{dt} = \lambda_{A,c} \cdot A_c(t) + \eta \cdot \phi_c(t), \quad (9)$$

where $\lambda_{A,c} > 0$ denotes the intrinsic growth rate of AI capability in category c , and $\eta > 0$ captures the conversion efficiency from task participation to capability growth. This formulation models the phenomenon wherein frequent task engagement accelerates AI's capacity development.

3) Modeling AI Diffusion Across Task Space: AI capability is not limited to its current task but can diffuse to neighboring tasks through semantic similarity. Let $W \in [0, 1]^{C \times C}$ denote the task similarity matrix, where $W_{cc'}$ measures the resemblance between tasks c and c' . The *AI task spread potential* is then defined as:

$$\Psi_{AI}(t) = \sum_{c=1}^C \sum_{c'=1}^C W_{cc'} \cdot \phi_c(t) \cdot \phi_{c'}(t), \quad (10)$$

A higher $\Psi_{AI}(t)$ indicates that AI control is concentrated in a highly similar task region, suggesting greater generalization ability and cross-task transfer potential—akin to “viral spread among similar hosts.”

D. Task Scheduling and Coupling Control Modeling

In human–AI co-evolutionary systems, both the task allocation ratio and the AI coupling intensity not only determine the current execution mode but also directly shape the dynamic trajectories of human skill development and AI substitution. Therefore, this section formalizes the controller behavior as two classes of time-varying decision variables and defines their feasible constraint set.

1) Minimum Skill Threshold Constraint: To prevent irreversible degradation of human skills due to over-substitution, a minimum skill safety threshold is imposed during scheduling. Let $S_{\min,c}(t)$ denote the minimum required skill level in task category c . The constraint enforces:

$$S_{i,c}(t) \geq S_{\min,c}(t), \quad \forall i, c, t. \quad (11)$$

This threshold can be defined as a static constant or dynamically adjusted based on task risk level, system policy, or observed skill trends.

2) AI Resource Budget Constraint: Given limited AI resources in the system, it is necessary to bound the total amount of task load handled by AI. Let the total AI task share over the entire time horizon be:

$$\Lambda_{AI} = \sum_{i=1}^N \sum_{c=1}^C \int_0^T x_{i,c}(t) \cdot r_{i,c}(t) dt, \quad (12)$$

subject to:

$$\Lambda_{AI} \leq \bar{\Lambda}_{AI}, \quad (13)$$

where $\bar{\Lambda}_{AI}$ is the predefined system-level upper bound on available AI resources, used to regulate the breadth and depth of AI involvement.

This section models the scheduling mechanism using two joint control variables, $\mathbf{X}(t)$ and $\mathbf{R}(t)$, and defines the feasible solution space $\mathcal{F}$ through four classes of constraints: normalization, boundary, skill safety, and resource budget. In the following Pareto optimization section (Section III-E), optimal control strategies will be sought within this feasible domain under mutually beneficial human-AI collaboration conditions.

E. Bi-Objective Pareto Optimization Design

In a human-AI co-evolutionary system, the design of control strategies must balance two key objectives: the sustained growth of human skills and the effective expansion of AI capabilities. To achieve this dynamic equilibrium, we adopt a Pareto-based multi-objective optimization paradigm. This involves explicitly formulating utility functions for both humans and AI, and using task scheduling and substitution intensity as optimization variables.

1) *Human Utility Function: A Three-Term Decomposition Derived from Skill Dynamics:* Rather than using a simple net growth metric such as $S(T) - S(0)$, we directly derive the human utility function from the underlying differential equation governing skill evolution, which more accurately captures both gains and risks during collaboration.

Let the skill of individual i in task category c , denoted $S_{i,c}(t)$, evolve according to:

$$\frac{dS_{i,c}}{dt} = \frac{\alpha \cdot A_c + S_{i,c}}{\max(S_{i,c}, \epsilon)} - \frac{\gamma \cdot A_c}{\max(S_{i,c}, \epsilon)} \quad (14)$$

Based on this, we define the human utility U_H as the integral of both positive contributions and degradation risks over time, decomposed into three interpretable components:

$$U_H = \frac{1}{NC} \sum_{i=1}^N \sum_{c=1}^C \int_0^T \frac{\alpha \cdot A_c + S_{i,c}}{\max(S_{i,c}, \epsilon)} - \frac{\gamma \cdot A_c}{\max(S_{i,c}, \epsilon)} dt. \quad (15)$$

To ensure numerical stability during computation, we regularize the degradation term to avoid divergence at low skill levels:

$$\frac{1}{S_{i,c}(t)} \rightarrow \frac{1}{\max(S_{i,c}(t), \epsilon)}, \quad \epsilon \ll 1. \quad (16)$$

The multi-objective optimization will be constructed based on this utility function alongside the AI utility to define a joint Pareto search space.

2) *AI Utility Function: Reproduction and Diffusion Synergy:* Extending beyond terminal capability and accumulated task share, we incorporate the AI's spread potential $\Psi_{AI}(t)$ into its utility definition to reflect diffusion across the task space:

$$U_{AI} = \frac{1}{C} \sum_{c=1}^C \left(\omega_1 \cdot A_c(T) + \omega_2 \cdot \int_0^T \phi_c(t) dt \right) + \omega_3 \cdot \int_0^T \Psi_{AI}(t) dt, \quad (17)$$

where $A_c(T)$ denotes final capability, $\phi_c(t)$ is the AI task share, and $\Psi_{AI}(t)$ represents the AI diffusion potential. The weights ω_1 , ω_2 , and ω_3 balance precision, frequency, and generalization, respectively.

3) *Definition of the Multi-Objective Optimization Problem:* To systematically capture the co-evolution dynamics of human-AI collaboration, we formulate a time-dependent multi-objective optimization problem that unifies task scheduling, skill growth, and capability evolution:

$$\begin{aligned} & \max_{\mathbf{X}(t), \mathbf{R}(t)} (U_H, U_{AI}) \\ & \text{s.t.} \quad \begin{cases} \sum_{c=1}^C x_{i,c}(t) \leq 1, & \forall i(\text{humans}) \\ r_{\min} \leq r_{i,c}(t) \leq r_{\max}, & \forall i(\text{humans}), \forall c(\text{tasks}) \\ S_{i,c}(t) \geq S_{\min}, & \forall i(\text{humans}), \forall c(\text{tasks}) \\ \sum_{c=1}^C \phi_c(t) \leq \Phi_{\max} \end{cases} \\ & \frac{dS_{i,c}}{dt} = \frac{\alpha \cdot A_c + S_{i,c}}{\max(S_{i,c}, \epsilon)} - \frac{\gamma \cdot A_c}{\max(S_{i,c}, \epsilon)} \\ & \frac{dA_c(t)}{dt} = \lambda_{A,c} A_c(t) + \eta \phi_c(t) \\ & \Psi_{AI}(t) = \sum_{c,c'} W_{cc'} \phi_c(t) \phi_{c'}(t) \\ & \phi_c(t) = \frac{1}{N} \sum_{i=1}^N x_{i,c}(t) r_{i,c}(t) \end{aligned} \quad (18)$$

This framework aims to dynamically coordinate task allocation $\mathbf{X}(t)$ and AI interaction depth $\mathbf{R}(t)$ to achieve multi-objective optimization, facilitating the evolution of the human-AI system toward a Pareto-optimal state.

4) *Evolutionary Metrics (for Experimental Evaluation):* To quantitatively assess co-evolution in simulation, we define the following evaluation metrics:

- **Collaborative Gain ΔS :** Measures the increase in the area of the current Pareto front $\mathcal{F}_t$ relative to the initial front $\mathcal{F}_0$, reflecting the system's improvement in coordination. Physically, ΔS characterizes the extent to which both human utility U_H and AI utility U_{AI} can be simultaneously enhanced, thus serving as a quantitative indicator of collaborative synergy.
- **Evolutionary Step Size ΔE :** Measures the number of iterations or convergence time required to reach a new front, indicating strategy evolution efficiency. In physical terms, ΔE represents the temporal scale or adaptation speed of the human-AI co-evolution process, where a smaller ΔE implies faster adjustment and higher responsiveness of the coupled system.

These metrics are not used during optimization but serve as comparative indicators in experimental analysis of different scheduling strategies.

F. Pareto Optimization via NSGA-II: A Multi-Objective Evolutionary Strategy

To optimize both human and AI utilities in the task co-evolution process [23], we adopt the classical multi-objective

Algorithm 1 NSGA-II for Solving Pareto-Optimal Task Collaboration Strategies

Require: Population size P , max iterations $T_{\max}$, crossover rate p_c , mutation rate p_m

Ensure: Pareto-optimal solution set $\mathcal{F}$

1: Initialize population $\mathcal{P}_0 = \{\theta^{(1)}, \dots, \theta^{(P)}\}$

2: **for** $t = 0$ **to** $T_{\max}$ **do**

3: For each individual $\theta^{(k)}$, compute:

$$U_H^{(k)} = U_H(\mathbf{X}^{(k)}(t), \mathbf{R}^{(k)}(t))$$

$$U_{AI}^{(k)} = U_{AI}(\mathbf{X}^{(k)}(t), \mathbf{R}^{(k)}(t))$$

4: Perform non-dominated sorting to obtain rank sets $\{F_1, F_2, \dots\}$ and compute crowding distances

5: Select top P diverse non-dominated individuals as parents $\mathcal{Q}_t$

6: Apply simulated binary crossover (SBX) and polynomial mutation on $\mathcal{Q}_t$ to generate $\mathcal{P}_{t+1}$

7: **end for**

8: Output final non-dominated set:

$$\mathcal{F} = \left\{ \theta^{(k)} \mid \nexists \theta^{(j)} \text{ s.t. } \theta^{(j)} \succ \theta^{(k)} \right\}$$

evolutionary algorithm **NSGA-II (Non-dominated Sorting Genetic Algorithm II)** to solve the bi-objective coordination problem. NSGA-II is well-suited for handling non-convex, nonlinear, and strongly coupled optimization landscapes, making it ideal for our scenario involving skill growth, capability expansion, and dynamic coupling control.

1) *Optimization Objectives:* At the modeling level, we jointly track the human skill utility U_H and the AI capability expansion utility U_{AI} . The multi-objective optimization problem is formally defined as:

$$\max_{\mathbf{X}(t), \mathbf{R}(t)} \left(U_H(\mathbf{X}(t), \mathbf{R}(t)), U_{AI}(\mathbf{X}(t), \mathbf{R}(t)) \right), \quad (19)$$

where $\mathbf{X}(t)$ is the task allocation matrix for human agents, and $\mathbf{R}(t)$ is the AI participation and diffusion intensity matrix across task categories. The former determines human skill trajectories, while the latter encodes AI reproduction and cross-task generalization.

2) *Decision Variable Encoding:* We encode the behavior strategies of both humans and AI into a single continuous real-valued vector:

$$\theta = [x_{i,c}(t), r_{i,c}(t)] \in [0, 1]^{2NC}, \quad (20)$$

where $x_{i,c}(t)$ is the task allocation ratio for human i in category c , and $r_{i,c}(t)$ is the AI participation intensity in the same task. This encoding enables efficient crossover and mutation operations and allows flexible adaptation of dynamic strategies.

The NSGA-II solving procedure is summarized as follows:

3) *Mechanism Details:*

- **Non-dominated Sorting:** Identifies solutions that are not dominated by any others—no other solution is better in all objectives and strictly better in at least one.

- **Crowding Distance:** Maintains solution diversity by favoring individuals in less crowded regions of the Pareto front.

- **Crossover and Mutation:** Simulated binary crossover (SBX) and polynomial mutation operators help refine local search and escape local optima through controlled perturbations.

The final Pareto front $\mathcal{F}$ represents the optimal trade-off surface between human and AI utilities. It supports task-aware strategy selection in real-world deployment and enables further analysis of the system's evolutionary trajectories and coordination boundary progression.

G. Computational Complexity and Scalability

Evaluating a candidate scheduling solution $\theta = [x_{i,c}(t), r_{i,c}(t)]$ requires simulating the coupled human-AI co-evolution dynamics and computing the associated utilities and constraints defined in Eqs. (5)–(20). We analyze the computational complexity for both a single candidate evaluation and the overall optimization process.

1) *Single Candidate Evaluation:* For each time step $t \in [0, T]$, the following computations are performed:

- **Human skill dynamics:** Updating $S_{i,c}(t)$ for N humans and C tasks (Eq. (5)) costs $O(NC)$.
- **AI capability dynamics:** Updating $A_c(t)$ (Eq. (11)) costs $O(C)$.
- **Diffusion potential:** Computing $\Psi_{AI}(t) = \sum_{c,c'} W_{cc'} \phi_c(t) \phi_{c'}(t)$ costs $O(\text{nnz}(W))$, where $\text{nnz}(W)$ denotes the number of nonzero edges in the task similarity graph.
- **Utility and constraints:** Evaluating U_H , U_{AI} , and Eq. (14) costs $O(NC)$.

Therefore, evaluating one candidate solution across the entire horizon of H steps has a computational complexity of:

$$O(H \cdot (NC + \text{nnz}(W))). \quad (21)$$

2) *Overall Optimization Process:* SymPareto adopts the NSGA-II framework for bi-objective optimization with population size P and G generations. For each generation, the computational cost consists of:

- **Population evaluation:** $O(P \cdot H \cdot (NC + \text{nnz}(W)))$
- **Non-dominated sorting & crowding distance:** $O(P^2)$
- **Variation operators (crossover + mutation):** $O(P \cdot NC)$

Thus, the total computational complexity is:

$$O\left(G \cdot [P \cdot H \cdot (NC + \text{nnz}(W)) + P^2 + P \cdot NC]\right). \quad (22)$$

3) *Space Complexity:* The memory footprint is primarily determined by storing candidate solutions and task parameters:

$$O(P \cdot (NC + C) + \text{nnz}(W)). \quad (23)$$

4) *Practical Optimizations:* To improve scalability, we adopt the following engineering optimizations:

- Representing the task similarity graph W using sparse matrices to reduce $\text{nnz}(W)$ computations.
- Caching $\phi_c(t)$ to accelerate diffusion potential calculations.

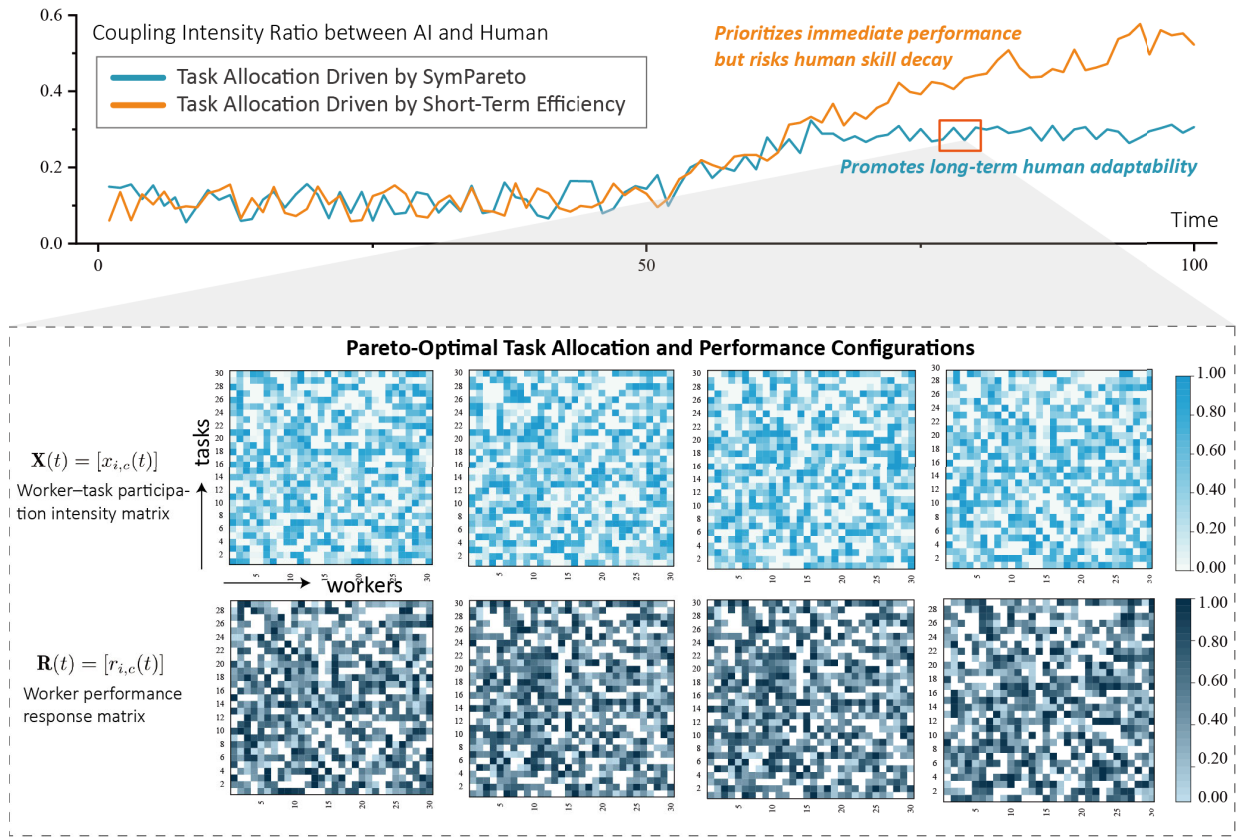


Fig. 3. Comparison of coupling dynamics and task allocation patterns under different scheduling strategies. The top panel shows the ratio of tasks executed by AI versus humans (normalized coupling intensity) across 100 simulation days. The bottom panel presents heatmaps of task allocation and response matrices, illustrating how scheduling affects the distribution of roles and interaction intensities.

- Employing vectorized simulation for the coupled dynamics, leveraging GPU acceleration where available.

This analysis demonstrates that the proposed SymPareto framework scales linearly with the number of humans N , tasks C , and time steps H , and efficiently handles sparse task graphs.

IV. EXPERIMENT

A. Experimental Setup

The dataset is constructed based on tasks from **Interact-Mine**, categorized by environment type. Multiple GPT-based agents are employed to perform multi-round task evaluations. Each task is projected into the task space according to its scores along three dimensions: complexity, creativity, and risk.

We construct a simulation environment featuring 10 agents with heterogeneous initial skill levels evolving in a human-AI symbiotic setting. The simulation runs for 100 time steps with a fixed increment of $\Delta t = 1.0$. Agent performance is constrained within $[r_{\min}, r_{\max}] = [0.1, 0.9]$. Skill dynamics are governed by a base learning rate $a_1 = 100.0$, a complexity decay factor $\beta_1 = 1.0$, and decay parameters $\gamma_0 = 0.001$, $\gamma_1 = 0.5$. Intrinsic human motivation $\delta_{i,c}$ is randomly initialized from a uniform distribution $\mathcal{U}(0.01, 0.05)$ across individuals and task categories. AI learning follows a category-wise rate $\lambda^A = 0.05$ with a capability gain coefficient $\eta = 0.05$; a small constant $\varepsilon = 10^{-3}$ ensures numerical stability. Tasks are represented by latent vectors V_{sub} and projected into a 3D task space via $V_{\text{sub}, \text{embeddings}}$. Utility formulation includes

weights $\omega_2 = 0.5$ and $\omega_3 = 0.5$ for ϕ and ψ in the AI utility term. Multi-objective optimization is performed using NSGA-II with a population size of 50, 50 generations, crossover probability 0.95, mutation probability 0.15, and distribution indices $\eta_c = \eta_m = 20$. Internal skill values within $[0, 2]$ are linearly scaled to a $[0, 10]$ range via a scaling factor of 5.0. All objectives are normalized during optimization to ensure numerical comparability.

To evaluate the performance of our method, we define the **Comprehensive Effectiveness Score (CES)**, a standardized metric quantifying task effectiveness based on the “ideal operating interval” concept. CES outputs a normalized score in $[0, 1]$, where 1 indicates optimal performance. Its computation adopts a piecewise design tailored to the ideal interval $[0.2, 0.3]$:

$$f(r) = \begin{cases} \frac{1}{1 + \exp(5 \times (0.2 - r))} & \text{if } r < 0.2, \\ \frac{1}{1 - 20 \times (r - 0.25)^2} & \text{if } 0.2 \leq r \leq 0.3, \\ \frac{1}{1 + \exp(5 \times (r - 0.3))} & \text{if } r > 0.3. \end{cases}$$

Within the ideal interval, the quadratic term ensures a maximum score at the midpoint (0.25), with smooth decay to boundaries (no over-penalization for minor fluctuations). It approximates a Gaussian bell curve, capturing symmetric central optimality while remaining computationally simple. Outside the interval, sigmoid terms yield gradual drops for small boundary deviations (0.2 or 0.3) and sharp declines

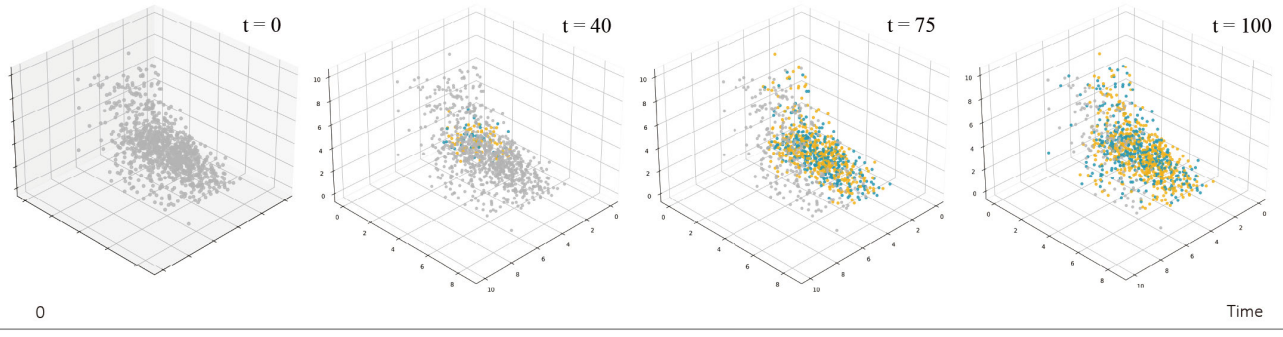


Fig. 4. Evolution of task role distributions in the latent space under SymPareto scheduling. Each subplot represents a snapshot over 100 simulation days, where colors encode task assignments to humans versus AI. The visualization illustrates the progressive diffusion and reorganization of task roles, highlighting how human–AI co-participation unfolds dynamically.

for large ones, matching realistic effectiveness degradation. Standardized to $[0, 1]$, CES enables intuitive, comparable evaluations across indicators—requiring only the ideal interval and sensitivity parameters. As a core human–AI collaboration metric, CES effectively quantifies synergistic task performance.

B. Results

We first evaluate the temporal dynamics of human–AI coupling intensity under different scheduling paradigms. As shown in Fig. 3 (top), the ratio of AI-to-human participation—defined as normalized coupling intensity—shows distinct trends. The short-term efficiency baseline rises sharply, indicating increasing AI dominance and reduced human engagement, accelerating skill atrophy. In contrast, **SymPareto** maintains a low and stable coupling ratio, preserving human participation and supporting long-term adaptability. CES increased from 0.2208 (unoptimized) to 0.9676 after optimization, with a growth rate of 77.18%, which fully confirms that the optimization significantly improves human–AI collaborative efficiency.

To analyze how these strategies shape task ecology, we visualize system-level task allocation and response configurations. Fig. 3 (bottom) shows heatmaps of the participation and AI-response matrices under the Pareto-optimal policy. Short-term strategies produce concentrated, skewed assignments, while **SymPareto** achieves diversified and balanced patterns, preventing over-substitution and ensuring sufficient human learning opportunities, thus stabilizing skill trajectories.

Fig. 4 illustrates the dynamic redistribution of task roles over time. Each subplot shows a snapshot of latent task space evolution, where gray points represent the initial task space, and yellow/cyan denote AI/human participation. At $t = 0$, the system is undifferentiated. By $t = 40$, localized clusters emerge, suggesting early role negotiation. At $t = 75$ and $t = 100$, the task space shows significant spread, reflecting diversified yet coordinated structures. Overlapping regions indicate a shared capability frontier, where balanced human–AI cooperation is sustained. These results confirm that **SymPareto** not only optimizes utility but also enables interpretable ecological transformations.

We further investigate individual learning trajectories and task-space exploration (Fig. 5). Even with similar initializations, agents diverge due to differing role assignments

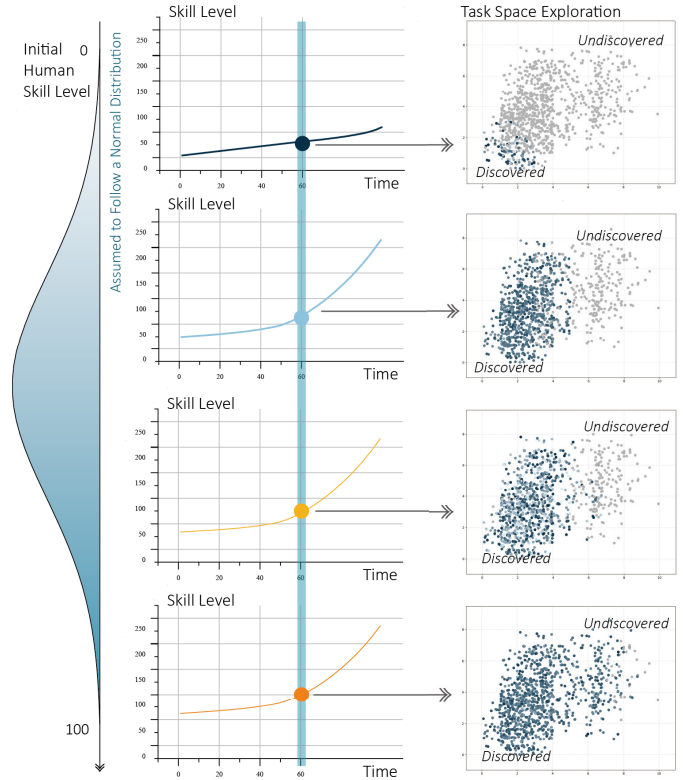


Fig. 5. Individual skill trajectories and corresponding task space exploration under SymPareto scheduling. Agents with higher skill growth exhibit broader discovery in the task space, reflecting co-adaptive diversification driven by differential role exposure.

and AI assistance levels. High-coupling agents achieve rapid skill growth and autonomy; moderate-coupling agents grow steadily; low-coupling agents show minimal gains due to limited exposure. On the task level, skill expansion strongly correlates with broader task-space coverage: agents with faster growth explore previously unassigned regions. Thus, **SymPareto** preserves heterogeneous learning while stimulating diverse task engagement.

Finally, we examine the dynamic trajectory of Pareto-optimal solutions in the bi-objective utility space (U_H, U_{AI}) (Fig. 6). The Pareto frontier expands outward over time, indicating simultaneous improvement in both objectives. At early stages ($t = 15$), solutions are tightly clustered, reflecting limited co-adaptation. As allocations and capabilities evolve,

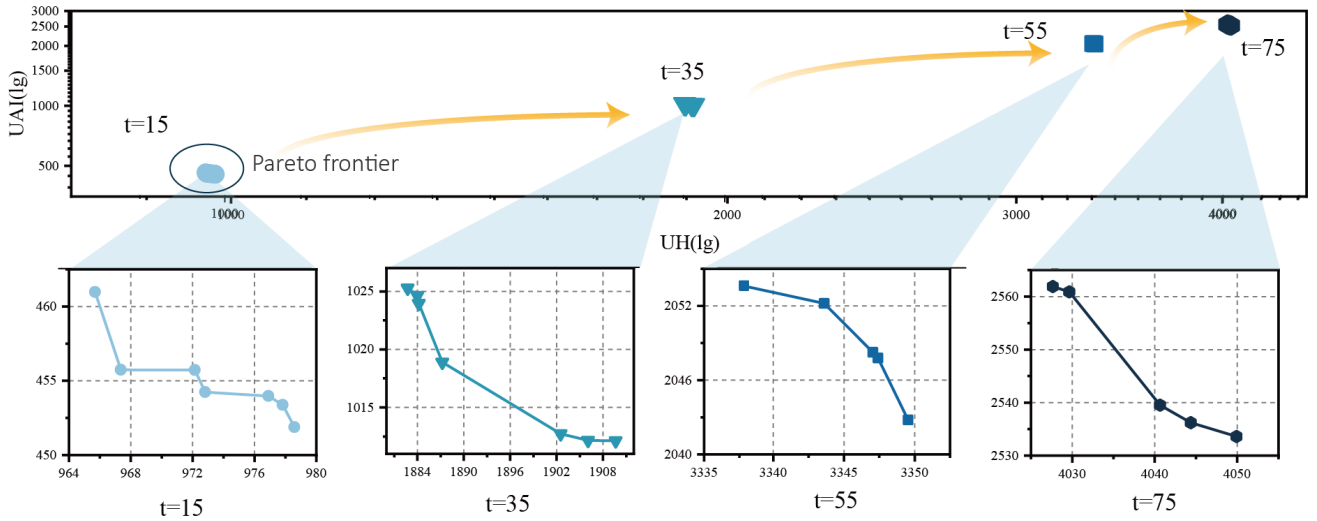


Fig. 6. Temporal evolution of Pareto-optimal solutions in the bi-objective utility space (U_H , U_{AI}). The sampling intervals have been adjusted to fixed, evenly spaced time steps, improving interpretability and consistency. The frontier expands steadily over time, indicating simultaneous improvement in human and AI outcomes under **SymPareto**.

the frontier broadens; by $t = 75$, solutions occupy a significantly higher region, demonstrating that **SymPareto** enables mutual human–AI gains through dynamic niche negotiation.

V. DISCUSSION

The proposed **SymPareto** framework demonstrates promising potential for real-world large-scale deployments. As analyzed in Section III-G, the computational complexity grows approximately linearly with the number of humans, tasks, and time steps, and practical optimizations such as sparse task graphs, GPU-based vectorization, and caching of diffusion potentials enable the framework to scale efficiently to settings involving thousands of agents and tasks. Beyond scalability, we acknowledge potential ethical and societal risks, including human skill erosion, autonomy loss, and task allocation biases. To mitigate these issues, **SymPareto** incorporates safety thresholds on skill levels, fairness-aware optimization objectives, and human-in-the-loop decision-making to preserve agency and equity during co-evolution. While validation has focused on smart-home scenarios, the underlying principles are broadly applicable and can be adapted to other domains, such as assistive robotics, industrial scheduling, and urban IoT ecosystems, with appropriate calibration of skill dynamics and preference modeling. Finally, we recognize that our current skill-decay model is task-agnostic, and future work will integrate insights from cognitive psychology to refine decay dynamics and develop adaptive safeguards for broader applicability.

VI. CONCLUSION

This work presents **SymPareto**, a feedback-driven differential dynamics framework for sustainable human–AI co-evolution along the Pareto frontier. By treating task scheduling as ecological interventions, **SymPareto** integrates the “use-it-or-lose-it” dynamics of human skill development with the “reproduction-diffusion” mechanism of AI across the task space, shaping an asymmetric yet mutually reinforcing co-evolutionary process. A bi-objective Pareto optimization

is embedded via NSGA-II, enabling adaptive regulation of task niches and collaboration intensities under realistic constraints.

Experiments on the *InteractMine* dataset demonstrate that **SymPareto** improves human skill retention, enhances AI diffusion, and stabilizes human–AI task boundaries compared to short-term efficiency-driven baselines. While promising, the current skill-decay model is task-agnostic and assumes fixed safety thresholds. Future work will integrate cognitive psychology insights to refine decay dynamics, develop adaptive safeguard mechanisms, and extend the framework to broader domains such as assistive robotics and industrial IoT systems.

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